

IOT BASED ELECTRICAL VEHICLE BATTERY MANAGEMENT SYSTEM

A Project Report submitted
In partial fulfilment of the requirements
For the award of the degree of

BACHELOR OF TECHNOLOGY

In

ELECTRICAL AND ELECTRONICS ENGINEERING

SUBMITTED BY

P.GAYATHRI PUSHPA	- 20PA1A0282
M.KETHAN SAI PRANAY	- 20PA1A0257
U.MANIKYALA PAVAN	- 20PA1A0293
V.SURYA TEJA SRUJITH KUMAR	- 20PA1A02A1
S.RAGHU VARMA	- 21PA5A0228

Under the esteemed guidance of

Mr. P. NAVEEN, M.Tech

Associate Professor in EEE Dept.



DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

VISHNU INSTITUTE OF TECHNOLOGY(AUTONOMOUS)

(Approved by AICTE; Accredited by NBA & NAAC; permanently Affiliated to JNTUK)

Vishnupur, Bhimavaram-534202, West Godavari Dist., A.P

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Department of Electrical and Electronics Engineering



CERTIFICATE

This is to certify that **P.Gayathri Pushpa - 20PA1A0282 , M.Kethan Sai Pranay - 20PA1A0257, U.Manikyala Pavan - 20PA1A0293, V.Surya Teja Srujith Kumar - 20PA1A02A1, S.Raghu Varma - 21PA5A0228** of Final year B.Tech, EEE have completed the project work on **“IoT Based Electrical Vehicle Battery Management System”** in partial fulfilment for the award of the degree of ‘Bachelor of Technology’ in Electrical and Electronics Engineering (EEE) in Vishnu Institute of Technology(Autonomous), Bhimavaram. This is a bonafide record of the work done by them during the academic year 2023-2024.The results of this project have not been submitted to any other university or Institute for the award of any degree.

Mr. P.Naveen, M.Tech
(Assoc.Prof, EEE Dept.)

Dr.R.V.D.Rama Rao, Ph.D.
(Professor & Head, EEE Dept.)

EXTERNAL EXAMINER

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With Sincere Regards,

Project Associates

P.Gayathri Pushpa	- 20PA1A0282
M.Kethan Sai Pranay	- 20PA1A0257
U.Manikyala Pavan	- 20PA1A0293
V.Surya Teja Srujith Kumar	- 20PA1A02A1
S.Raghu Varma	- 21PA5A0228

ABSTRACT

An IoT-based battery management system (BMS) is a technology that uses the internet of things (IoT) to monitor and control batteries in various applications. The BMS consists of sensors, microcontrollers, communication modules, and cloud-based servers that work together to collect data, analyse it, and optimize battery usage.

Electric vehicle Battery Management Systems (BMS) play a crucial role in ensuring the safe and efficient operation of battery packs, thereby contributing to the performance and longevity of electric vehicles (EVs). This project focuses on the design, implementation, and evaluation of a sophisticated BMS solution tailored for electric vehicle applications. Leveraging advanced software algorithms, precise sensors, and intelligent control strategies, the BMS enables real-time monitoring, analysis, and optimization of critical battery parameters, including voltage, current, temperature, state of charge (SoC), and state of health (SoH). The software architecture orchestrates seamless communication between hardware components, facilitating accurate data acquisition and enabling proactive measures to prevent overcharging, over-discharging, and thermal runaway.

Results obtained from the implementation and evaluation of the BMS demonstrate its efficacy in enhancing battery management, improving operational safety, and maximizing vehicle performance. Through comprehensive analysis and validation, the project underscores the significance of software-driven solutions in advancing electric vehicle technology and shaping the future of sustainable mobility. Further research and development efforts are recommended to explore the integration of advanced algorithms, sensor technologies, and fault-tolerant systems, thereby enhancing the robustness and scalability of BMS solutions for diverse electric vehicle applications.

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CHAPTER – 1

INTRODUCTION

1.1 OBJECTIVE

BMS is used in many contemporary and commercial frameworks to improve battery activity and for evaluation in order to maintain the battery state, to the greatest extent possible, away from a disastrous condition and to lengthen battery lifespan. The battery's charge, temperature, and current are therefore checked using a variety of processes.

Electric vehicle Battery Management Systems (BMS) play a crucial role in monitoring and managing the health and performance of the battery pack. One of the aspects BMS systems focus on is ensuring safety, which includes measures to prevent overheating and potential fire hazards.

The project on Electric Vehicle Battery Management System (BMS) and Fire Protection represents a crucial advancement in the rapidly evolving landscape of electric vehicle (EV) technology. As the automotive industry pivots towards sustainable mobility solutions, electric vehicles have emerged as a promising alternative to traditional internal combustion engine vehicles. However, with the proliferation of EVs comes the imperative to ensure not only their performance but also their safety and reliability.

BMS continuously monitors the temperature, voltage, and state of charge of individual cells within the battery pack. This monitoring helps identify any anomalies or issues that could lead to overheating or potential fire hazards. Additionally, it includes circuitry to prevent overcurrent situations, which could lead to excessive heat generation and potential fire risks, by regulating the flow of current to and from the battery pack.

Furthermore, BMS may control the thermal management system of the battery pack, which includes features such as active cooling or heating to maintain optimal operating temperatures. Proper thermal management reduces the risk of thermal runaway and fire incidents. It also tracks the state of health of the battery pack over time, identifying degradation or damage that could increase the risk of fire hazards and implementing strategies to mitigate these risks, such as adjusting charging rates or limiting the operating conditions.

In the event of a critical fault or imminent danger, BMS can initiate safety shutdown procedures to disconnect the battery pack from the rest of the vehicle's systems, preventing potential fire incidents. Moreover, BMS communicates with the vehicle's onboard systems and may provide warnings or alerts in case of potential fire risks. It can also record and report relevant data to aid in post-incident analysis and preventive measures.

In addition to the BMS, electric vehicles typically incorporate various passive and active safety measures to mitigate fire risks, including robust battery pack enclosures, fire-resistant materials, and dedicated fire suppression systems. Overall, BMS plays a central role in fire protection by actively monitoring and managing the battery system's health and performance.

Moreover, the project extends its scope to encompass fire detection and suppression mechanisms, crucial for mitigating the risks associated with battery fires. By incorporating sensors capable of detecting smoke and elevated temperatures, coupled with rapid-response suppression systems, the project aims to provide a comprehensive safety net against fire incidents within EVs.

1.2 ORGANIZATION OF PROJECT

CHAPTER 1:

In this chapter, we will discuss about the main objective of the project

CHAPTER 2:

In this chapter, we will discuss about the literature survey

CHAPTER 3:

In this chapter, we will discuss about the hardware components of the project

CHAPTER 4:

In this chapter, we will discuss about the operation and results of this project

CHAPTER 5:

In this we will get the conclusion of the Battery Management System of an Electric vehicle

1.3 CONCLUSION

The conclusion of this chapter is the brief idea of this project and the objective of the project

CHAPTER – 2

LITERATURE REVIEW

2.1 INTRODUCTION

A comprehensive literature review on electric vehicle Battery Management Systems (BMS) encompasses various aspects, including the importance of BMS in EVs, key functionalities, existing technologies, and recent advancements.

2.2 LITERATURE REVIEW

Development of Electric Vehicle Battery Management System with Charge Balance Control

This paper presents the development of an Electric Vehicle Battery Management System (BMS) with a focus on charge balance control, crucial for maximizing efficiency and longevity of electric vehicle batteries. Incorporating advanced algorithms and sensing technologies, the BMS monitors individual cell states of charge (SoC) and health (SoH), ensuring uniform charging and discharging to mitigate degradation and capacity imbalance issues. Emphasizing scalability and flexibility, the system integrates seamlessly into EV architectures, facilitating real-time data exchange and accommodating diverse battery chemistries and configurations. Experimental results demonstrate the BMS's efficacy in enhancing battery performance, prolonging lifespan, and optimizing energy utilization, thus contributing to the advancement of electric vehicle technology and promoting their widespread adoption as sustainable transportation solutions.

Real-time state-of-charge estimation for electric vehicle batteries using a coupled electro-chemical-thermal model

This paper presents a real-time state-of-charge (SoC) estimation method for electric vehicle (EV) batteries utilizing a coupled electrochemical-thermal model. By integrating both electrochemical and thermal dynamics, the proposed model offers improved accuracy in predicting battery SoC during dynamic operating conditions. Leveraging real-time measurements of battery voltage and temperature, coupled with an extended Kalman filter, the estimation algorithm accounts for variations in cell parameters and environmental factors. Experimental validation demonstrates the effectiveness of the approach, showcasing its ability to provide precise SoC estimation, crucial for optimizing battery management strategies and enhancing the overall performance and reliability of EVs.

Development of a Fire detection system for Lithium-ion battery in Electric Vehicles

This paper presents the development of a fire detection system tailored specifically for lithium-ion batteries in electric vehicles (EVs). With the growing popularity of EVs, concerns regarding battery safety, particularly the risk of thermal runaway and fire incidents, have become paramount. The proposed system integrates multiple sensors to monitor various parameters such as temperature, gas emissions, and current fluctuations, enabling early detection of potential fire hazards. Utilizing advanced signal processing techniques and machine learning algorithms, the system analyzes sensor data in real-time to identify abnormal patterns indicative of impending thermal runaway events. Experimental validation under simulated conditions demonstrates the system's effectiveness in accurately detecting and mitigating fire risks, offering a proactive solution to enhance the safety and reliability of EV battery systems.

An Integrated Thermal Management System for Lithium-ion battery pack in Electric Vehicles

This paper presents the design and implementation of an integrated thermal management system specifically tailored for lithium-ion battery packs in electric vehicles (EVs). Efficient thermal management is crucial for maintaining battery performance, extending lifespan, and ensuring safety. The proposed system integrates active cooling, passive cooling, and thermal insulation techniques to

effectively regulate battery temperature and minimize thermal gradients within the pack. Utilizing real-time monitoring and control strategies, the system dynamically adjusts cooling mechanisms based on battery operating conditions, environmental factors, and vehicle usage patterns. Experimental validation under various driving scenarios demonstrates the system's ability to optimize battery thermal performance, enhance energy efficiency, and mitigate the risk of thermal runaway events, thereby contributing to the overall reliability and longevity of EV battery systems.

Here's a detailed review covering these aspects:

Importance of BMS in Electric Vehicles

Electric vehicles (EVs) rely on battery packs as their primary energy storage system. A BMS is essential for managing these batteries to ensure optimal performance, safety, and longevity. According to research by Plett et al. (2019), BMS plays a critical role in monitoring battery health, controlling charging and discharging processes, and protecting against overcharging, over-discharging, and thermal runaway. Without an effective BMS, the reliability and safety of EVs would be compromised.

Key Functionalities of BMS

The functionalities of a BMS typically include state of charge (SoC) estimation, state of health (SoH) monitoring, cell balancing, thermal management, and fault diagnosis. According to Zhang et al. (2017), SoC estimation methods range from simple voltage-based approaches to sophisticated algorithms incorporating current integration and model-based techniques. SoH monitoring involves tracking battery degradation over time, while cell balancing ensures uniform energy distribution among battery cells to maximize pack capacity and lifespan.

Existing Technologies and Architectures

Various BMS architectures and technologies have been proposed and implemented in commercial EVs. Traditional BMS designs often rely on centralized architectures, where a single control unit manages the entire battery pack. However, recent research by Liu et al. (2020) explores distributed BMS architectures, where individual cell modules or clusters communicate with a central controller, offering scalability, redundancy, and improved fault tolerance.

Recent Advancements and Research Trends

Recent advancements in BMS technology focus on enhancing accuracy, efficiency, and reliability. Machine learning and artificial intelligence techniques are increasingly being applied to BMS for improved SoC estimation, fault detection, and predictive maintenance. Additionally, research by Hu et al. (2021) highlights the integration of advanced sensor technologies, such as fiber optic sensors and wireless sensor networks, for real-time monitoring of battery parameters and environmental conditions.

Conclusion

In conclusion, the literature underscores the critical importance of BMS in electric vehicles, emphasizing its role in ensuring battery performance, safety, and longevity. As EV technology continues to evolve, ongoing research efforts are focused on advancing BMS functionalities, exploring new architectures, and integrating emerging technologies to address the evolving needs of the electric vehicle industry.

CHAPTER – 3

3.1 INTRODUCTION

This section delves into the design and implementation of an EV BMS project, aiming to provide a comprehensive understanding of its components and functionalities.

Beginning with an overview of the project's block diagram, we will explore the key components comprising the BMS architecture and elucidate their roles in safeguarding battery health, optimizing performance, and ensuring the reliability of the electric vehicle's power source.

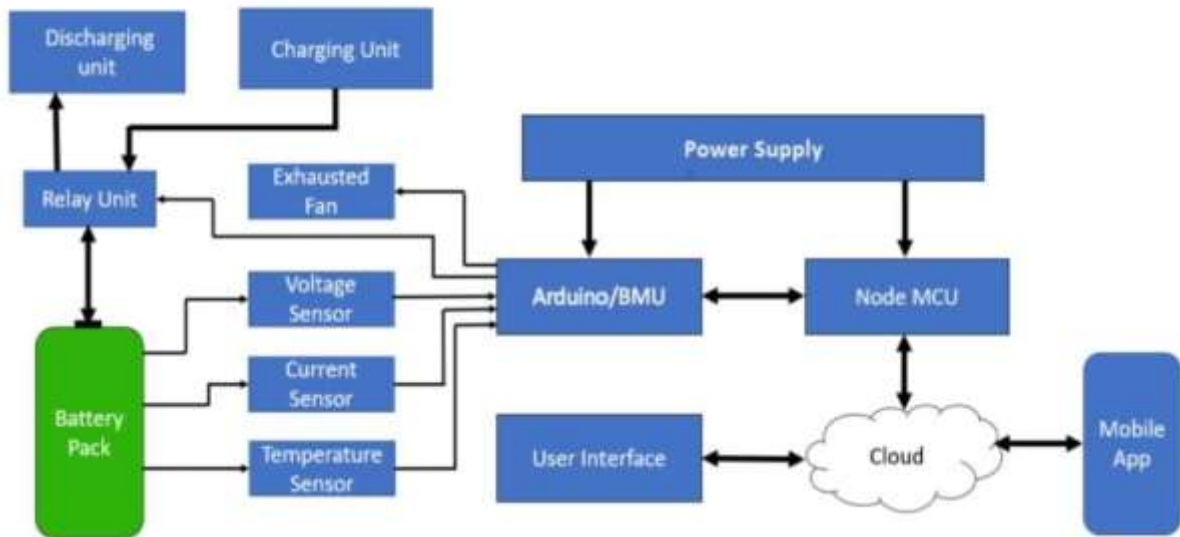


Fig 3.1 Block Diagram

Battery Pack: This is the primary power source for the electric vehicle. It consists of multiple lithium-ion battery cells connected in series and/or parallel configurations.

Battery Management System (BMS):

- **Microcontroller Unit (MCU):** Controls the overall operation of the BMS. It interfaces with various sensors and control modules.
- **Cell Voltage Sensors:** These sensors monitor the voltage of each individual battery cell in the pack.
- **Temperature Sensors:** Measure the temperature of the battery cells to ensure they are within safe operating limits.
- **Current Sensor:** Monitors the current flowing into and out of the battery pack.
- **BMS Control Module:** Receives data from sensors and issues commands to manage cell balancing, charging, and discharging.
- **Communication Interface:** Allows the BMS to communicate with other vehicle systems or external devices for monitoring and control.

3.2 COMPONENTS USED

- 1.Arduino Nano
- 2.Node MCU
- 3.Relay(5v)
- 4.Voltage Sensor (0-25vdc)

5.Current Sensor(30amp) (ACS712)

7.Temperature Sensor (NTC 103)

8.Transformer

9.Bridge Rectifier

10.Dc Motor

11. 1.5ohm 10w Resistor

3.3 POWER SUPPLY

A power supply (sometimes known as a power supply unit or PSU) is a device or system that supplies electrical or other types of energy to an output load or group of loads. The term is most commonly applied to electrical energy supplies, less often to mechanical ones, and rarely to others.

All digital circuits work only with low DC voltage. A power supply unit is required to provide the appropriate voltage supply. This unit consists of transformer, rectifier, filter and a regulator. AC voltage typically of 230Vrms is connected to a transformer which steps that AC voltage down to the desired AC voltage level. A diode rectifier then provides a full wave rectified voltage that is initially filtered by a simple capacitor filter to produce a DC voltage. This resulting DC voltage usually has some ripple or AC voltage variations. Regulator circuit can use this DC input to provide DC voltage that not only has much less ripple voltage but also remains in the same DC value, even when the DC voltage varies, or the load connected to the output DC voltage changes. The required DC supply is obtained from the available AC supply after rectification, filtration and regulation.

The main components used in the power supply unit are Transformer, Rectifier, Filter and Regulator. The 230V AC supply is converted into 12V AC supply through the transformer. The output of the transformer has the same frequency as in the input AC power. This AC power is converted into DC power through diodes. Here the bridge diode is used to convert AC supply to the DC power supply. This converted DC power supply has the ripple content and for normal operation of the circuit, the ripple content of the DC power supply should be as low as possible. Because the ripple content of the power supply will reduce the life of the circuit. So, to reduce the ripple content of the DC power supply, the large value of capacitance filter is used.

This filtered output will not be the regulated voltage. For this purpose, IC7805 regulator IC is used in the circuit.

3.4 TRANSFORMER

3.4.1 Introduction

Transformer is a device that transfers electrical energy from one circuit to another through inductively coupled conductors — the transformer's coils or "windings". Except for air-core transformers, the conductors are commonly wound around a single iron-rich core, or around separate but magnetically-coupled cores. A varying current in the first or "primary" winding creates a varying magnetic field in the core (or cores) of the transformer. This varying magnetic field induces a varying electromotive force (EMF) or "voltage" in the "secondary" winding. This effect is called mutual induction.



Fig 3.2 Transformer

If a load is connected to the secondary circuit, electric charge will flow in the secondary winding of the transformer and transfer energy from the primary circuit to the load connected in the secondary circuit.

The secondary induced voltage V_S , of an ideal transformer, is scaled from the primary V_P by a factor equal to the ratio of the number of turns of wire in their respective windings:

$$\frac{V_S}{V_P} = \frac{N_S}{N_P}$$

By appropriate selection of the numbers of turns, a transformer thus allows an alternating voltage to be stepped up — by making

N_S more than N_P — or stepped down, by making it

3.4.2 Basic Parts of Transformer

In its most basic form a transformer consists of:

1. A primary coil or winding.
2. A secondary coil or winding.
3. A core that supports the coils or windings.

Refer to the transformer circuit in figure as you read the following explanation: The primary winding is connected to a 60-hertz ac voltage source. The magnetic field (flux) builds up (expands) and collapses (contracts) about the primary winding. The expanding and contracting magnetic field around the primary winding cuts the secondary winding and induces an alternating voltage into the winding. This voltage causes alternating current to flow stepped up or down depending on the design of the primary and secondary windings.

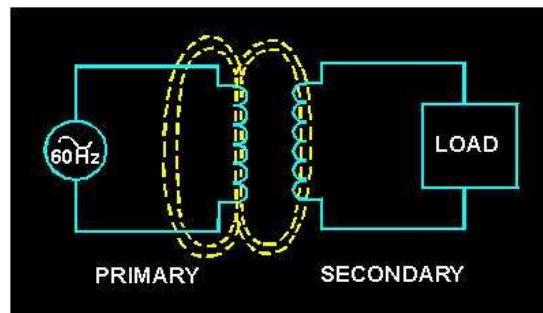


Fig 3.3 Primary and secondary Windings

3.4.3 The Components of Transformer

Two coils of wire (called windings) are wound on some type of core material. In some cases the coils of wire are wound on a cylindrical or rectangular cardboard form. In effect, the core material is air and the transformer is called an AIR-CORE TRANSFORMER. Transformers used at low frequencies, such as 60 hertz and 400 hertz, require a core of low-reluctance magnetic material, usually iron. This type of transformer is called an IRON-CORE TRANSFORMER. Most power transformers are of the iron-core type. The principle parts of a transformer and their functions are:

- The Core, which provides a path for the magnetic lines of flux.
- The Primary Winding, which receives energy from the ac source.
- The Secondary Winding, which receives energy from the primary winding and delivers it to the load.
- The Enclosure, which protects the above components from dirt, moisture, and mechanical damage.

3.5 Bridge Rectifier

3.5.1 Introduction

In an electric vehicle Battery Management System (BMS), a bridge rectifier serves as a crucial component for converting alternating current (AC) to direct current (DC), ensuring stable and consistent power supply to various subsystems within the BMS. Here's a description of the bridge rectifier's role within the BMS project:



Fig 3.4 Bridge Rectifier

AC to DC Conversion: The primary function of the bridge rectifier is to convert the alternating current (AC) output from the charging source, such as the vehicle's onboard charger or external charging station, into direct current (DC). This conversion is essential as most electronic components within the BMS operate on DC power.

Charging System Integration: The bridge rectifier facilitates seamless integration of the charging system with the BMS by converting the AC voltage from the charger into a form compatible with the BMS components. This ensures that the battery pack receives the correct DC voltage for charging and monitoring purposes.

Voltage Regulation: Bridge rectifiers often include circuitry for voltage regulation, ensuring that the output DC voltage remains within specified limits regardless of fluctuations in the input AC voltage. This regulation is crucial for maintaining the stability and reliability of the BMS components, especially during charging cycles.

Protection Features: Modern bridge rectifiers may incorporate built-in protection features such as overcurrent protection, overvoltage protection, and reverse polarity protection. These features help safeguard the rectifier and the connected components from potential damage due to excessive currents, voltages, or incorrect polarity.

Efficiency Optimization: Bridge rectifiers are designed to operate with high efficiency, minimizing energy losses during the AC to DC conversion process. This efficiency optimization is essential for maximizing the overall energy efficiency of the BMS and reducing wasted energy during charging and discharging cycles.

Compact Design: Bridge rectifiers are available in compact and lightweight designs, making them suitable for integration into the space-constrained environment of electric vehicles. Their small size and high power density contribute to the overall efficiency and performance of the BMS system.

Overall, the bridge rectifier plays a critical role in ensuring reliable AC to DC conversion, voltage regulation, and protection within the electric vehicle BMS. By providing stable and consistent DC power supply to the BMS components, it contributes to the efficient operation and longevity of the battery system.

3.6 BUCK CONVERTER

In the design and implementation of the Electric Vehicle Battery Management System (BMS), a buck converter serves as an essential component to regulate the voltage supplied to different parts of the system. The buck converter is a type of DC-DC converter that steps down the voltage from the battery pack to a level suitable for various components within the BMS.



Fig 3.5 Buck Converter

Functionality: The buck converter works by efficiently converting a higher input voltage from the battery pack to a lower output voltage required by certain components of the BMS. This is achieved through the use of switching elements, such as transistors, capacitors, and inductors, which control the flow of current and voltage levels.

Operation: During operation, the buck converter periodically switches the input voltage across an inductor, storing energy in the inductor during the on-time and releasing it to the load during the off-

time. By controlling the duty cycle of the switching signal, the output voltage can be adjusted as needed.

Efficiency: One of the key advantages of using a buck converter is its high efficiency, which minimizes power losses during voltage conversion. This is particularly important in electric vehicles where maximizing energy efficiency is crucial for extending driving range and optimizing battery performance.

Protection Features: The buck converter may also incorporate protection features such as overvoltage protection, overcurrent protection, and thermal shutdown to safeguard the system against faults and ensure reliable operation under various operating conditions.

Integration with BMS: The buck converter is integrated into the overall BMS architecture to provide stable and regulated power supply to components such as sensors, microcontrollers, communication modules, and other subsystems involved in battery monitoring, balancing, and control.

Optimization: The design of the buck converter is optimized to meet the specific voltage requirements and power constraints of the BMS, taking into account factors such as load variation, efficiency targets, size constraints, and thermal management considerations.

3.7 ARDUINO NANO

3.7.1 Introduction

The Arduino Nano is a small Arduino board based on ATmega328P or ATmega628 Microcontroller. The connectivity is the same as the Arduino UNO board.

The Nano board is defined as a sustainable, small, consistent, and flexible microcontroller board. It is small in size compared to the UNO board. The Arduino Nano is organized using the Arduino (IDE), which can run on various platforms. Here, IDE stands for Integrated Development Environment.

The devices required to start our projects using the Arduino Nano board are Arduino IDE and mini USB. The Arduino IDE software must be installed on our respected laptop or desktop. The mini USB transfers the code from the computer to the Arduino Nano board.

3.7.2 Drawback

The DC power jack is absent in Nano. Thus, we cannot use a battery to apply any external power supply.

The Arduino Nano is shown below:

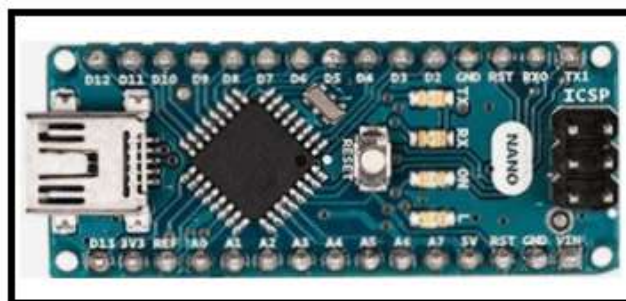


Fig 3.6 Arduino Nano

3.7.3 Arduino Nano Pinout

Arduino nano pin configuration is shown in fig 415 and each pin functionality is discussed below.

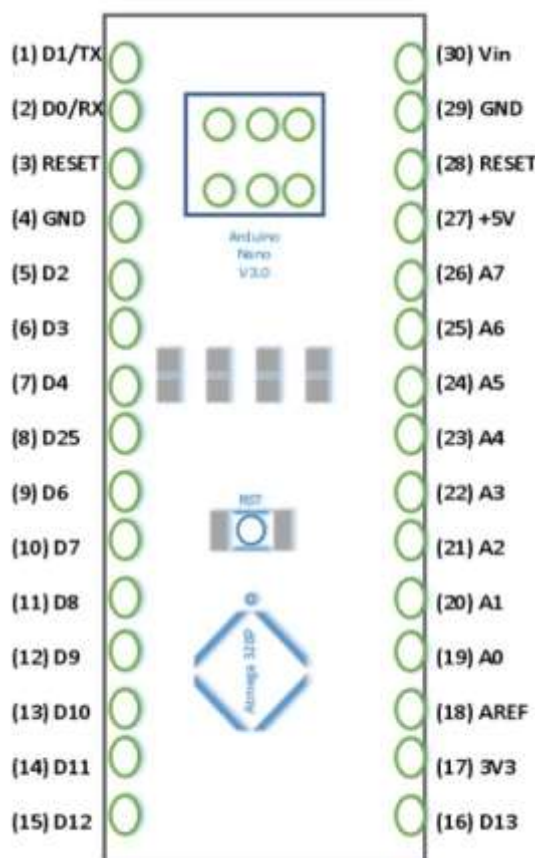


Fig 3.7 Arduino-nano-pinout

Power Pin (Vin, 3.3V, 5V, GND): These pins are power pins

- Vin is the input voltage of the board, and it is used when an external power source is used from 7V to 12V.
- 5V is the regulated power supply voltage of the nano board and it is used to give the supply to the board as well as components.
- 3.3V is the minimum voltage which is generated from the voltage regulator on the board.
- GND is the ground pin of the board

RST Pin(Reset): This pin is used to reset the microcontroller

Analog Pins (A0-A7): These pins are used to calculate the analog voltage of the board within the range of 0V to 5V

I/O Pins (Digital Pins from D0 – D13): These pins are used as an i/p otherwise o/p pins. 0V & 5V

Serial Pins (Tx, Rx): These pins are used to transmit & receive TTL serial data.

External Interrupts (2, 3): These pins are used to activate an interrupt.

PWM (3, 5, 6, 9, 11): These pins are used to provide 8-bit of PWM output.

SPI (10, 11, 12, & 13): These pins are used for supporting SPI communication.

Inbuilt LED (13): This pin is used to activate the LED.

IIC (A4, A5): These pins are used for supporting TWI communication.

AREF: This pin is used to give reference voltage to the input voltage

Why is Arduino Nano used?

Using the constant voltage, the Arduino Nano is used to **produce a clock of a precise frequency.**

3.7.4 Difference between Arduino UNO and Nano board?

- The Arduino Nano has a compact size and mini USB cable than the Arduino UNO. We can use Nano instead of UNO because both operate on the microcontroller ATmega328p.
- The Arduino UNO is also easily available than Nano. It is considered as the standard board available in the market, which is easy for use for the starters or beginners.
- The Nano is available in PDIP (Plastic Dual - Inline Package), while Arduino UNO is available in TQFP (Plastic Quad Flat Pack).
- The Arduino UNO includes **6 analog pin inputs, 14 digital pins, a USB connector, a power jack, and an ICSP (In-Circuit Serial Programming) header.** The Arduino Nano includes an I/O pin set of 14 digital pins and 8 analog pins. It also includes 6 Power pins and 2 Reset pins.

3.7.5 Memory

The memory in Arduino Nano is shown in the below figure:

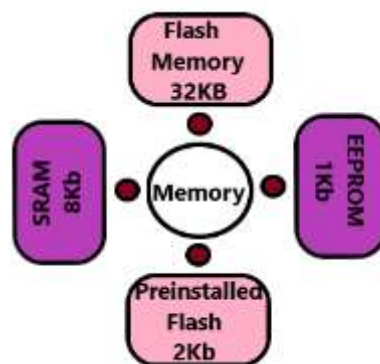


Fig 3.8 Memory in Arduino Nano

The preinstalled flash has a boot loader, which takes the memory of 2Kb.

Technical Specifications

The technical specifications of the Arduino Nano board are:

- The operating voltage of the Nano board varies from 5V to 12V.
- The total pins in Nano are 22 Input/Output pins.
- There are 14 digital pins and 8 analog pins.
- There are 6 PWM (Pulse Width Modulation) pins among the 14 digital pins. The 6 PWM pins in Arduino Nano are used to convert the digital signals into the analog signals. The conversion takes place by varying the width of the pulse.
- The crystal oscillator present in Arduino Nano comes with a frequency of 16MHz.

- The Arduino Nano is used in various applications such as **Robotics, Control System, Instrumentation, Automations, and Embedded Systems.**
- The projects created using Arduino Nano are **QR Code Scanner, DIY Arduino Pedometer, etc.**
- We can also connect Arduino Nano to the Wifi.
- The functionality of Nano is similar to the Arduino UNO.
- The flexibility and eco-friendly nature of Nano make it a unique choice to create electronic devices and projects with compact size.

3.8 Node MCU ESP8266

3.8.1 Description

NodeMCU is an open-source Lua based firmware and development board specially targeted for IoT based Applications. It includes firmware that runs on the ESP8266 Wi-Fi SoC from Espressif Systems, and hardware which is based on the ESP-12 module.

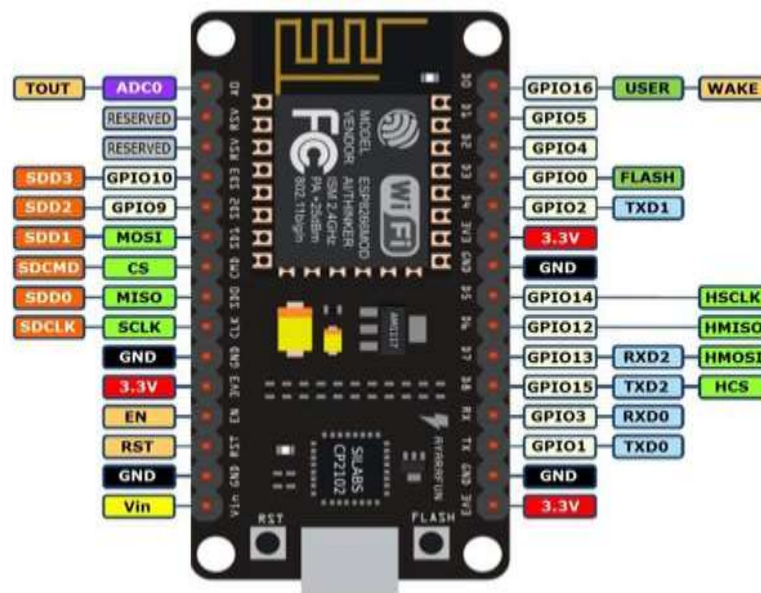


Fig 3.9 Node MCU ESP8266

3.8.2 NodeMCU ESP8266 Specifications & Features

- Microcontroller: Tensilica 32-bit RISC CPU Xtensa LX106
- Operating Voltage: 3.3V
- Input Voltage: 7-12V
- Digital I/O Pins (DIO): 16
- Analog Input Pins (ADC): 1
- UARTs: 1
- SPIs: 1
- I2Cs: 1
- Flash Memory: 4 MB

- SRAM: 64 KB
- Clock Speed: 80 MHz
- USB-TTL based on CP2102 is included onboard, Enabling Plug n Play
- PCB Antenna
- Small Sized module to fit smartly inside your IoT projects

3.8.3 Node MCU Development Board Pinout Configuration

Pin Category	Name	Description
Power	Micro-USB, 3.3V, GND, Vin	Micro-USB: NodeMCU can be powered through the USB port 3.3V : Regulated 3.3V can be supplied to this Pin to power the board GND : Ground Pins Vin : External Power Supply
Control Pins	EN, RST	The pin and the button resets the microcontroller.
Analog Pin	A0	Used to measure analog voltage in the range of 0-3.3V
GPIO Pins	GPIO1 to GPIO16	NodeMCU has 16 general purpose input-output pins on its board
SPI Pins	SD1, CMD, SD0, CLK	NodeMCU has four pins available for SPI communication
UART Pins	TXD0, RXD0, TXD2, RXD2	NodeMCU has two UART interfaces, UART0(RXD0 & TXD0) and UART1(RXD1 & TXD1). UART1 is used to upload the firmware/program.
I2C Pins		NodeMCU has I2C functionality support but due to the internal functionality of these pins, you have to find which pin is I2C.

Table 3.1 Node MCU Development Board Pinout Configuration

3.8.4 Programming Node MCU ESP8266 with Arduino IDE

The NodeMCU Development Board can be easily programmed with Arduino IDE since it is easy to use. Programming NodeMCU with the Arduino IDE will hardly take 5-10 minutes. All you need is the Arduino IDE, a USB cable and the NodeMCU board itself.

3.8.5 Brief About Node MCU ESP8266

The NodeMCU ESP8266 development board comes with the ESP-12E module containing the ESP8266 chip having Tensilica Xtensa 32-bit LX106 RISC microprocessor. This microprocessor supports RTOS and operates at 80MHz to 160 MHz adjustable clock frequency. NodeMCU has 128 KB RAM and 4MB of Flash memory to store data and programs. Its high processing power with in-built Wi-Fi / Bluetooth and Deep Sleep Operating features make it ideal for IoT projects.

NodeMCU can be powered using a Micro USB jack and VIN pin (External Supply Pin). It supports UART, SPI, and I2C interface.

3.9 Relay Switch



Fig 3.10 Relay Pin Out

Pin Number	Pin Name	Description
1	Coil End 1	Used to trigger(On/Off) the Relay, Normally one end is connected to 12V and the other end to ground
2	Coil End 2	Used to trigger(On/Off) the Relay, Normally one end is connected to 12V and the other end to ground
3	Common (COM)	Common is connected to one End of the Load that is to be controlled
4	Normally Close (NC)	The other end of the load is either connected to NO or NC. If connected to NC the load remains connected before trigger

Table 3.2 Pin Configuration

3.9.1 How to Use a Relay?

Relays are most commonly used switching device in electronics. There are two important parameters of relay, first is the Trigger Voltage, this is the voltage required to turn on the relay that is to change the contact from Common → NC to Common → NO. The other parameter is your Load Voltage & Current, this is the amount of voltage or current that the NC, NO or Common terminal of the relay could withstand, in our case for DC it is maximum of 30V and 10A. Make sure the load you are using falls into this range.

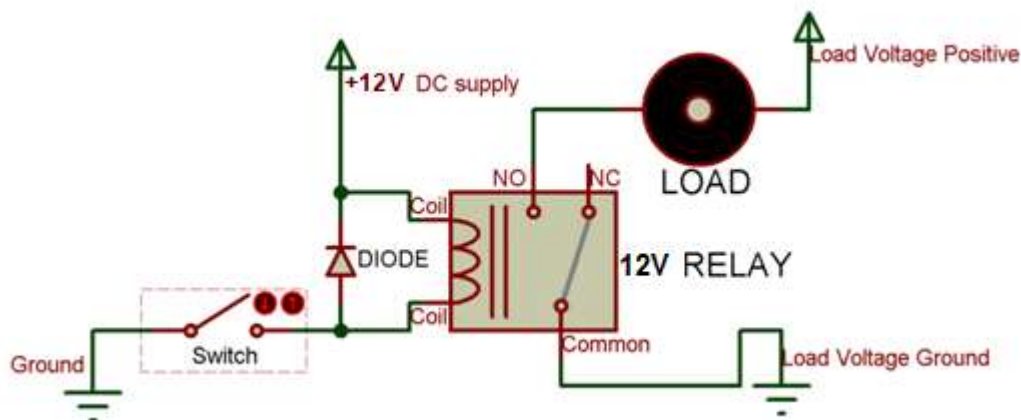


Fig 3.11 Relay Connection

The above diagram is for **relay triggering circuit**. Since the relay has 12V trigger voltage we have used a +12V DC supply to one end of the coil and the other end to ground through a switch. For switching we are using a transistor as a switching device. You can also notice a diode connected across the coil of the relay, this diode is called the Fly back Diode. The purpose of the diode is to protect the switch from high voltage spike that can produced by the relay coil. As shown one end of the load can be connected to the Common pin and the other end is either connected to NO or NC. If connected to NO the load remains disconnected before trigger and if connected to NC the load remains connected before triggered.

3.10 ELECTRIC BATTERY

An **electric battery** is a source of electric power consisting of one or more electrochemical cells with external connections for powering electrical devices. When a battery is supplying power, its positive terminal is the cathode and its negative terminal is the anode. The terminal marked negative is the source of electrons that will flow through an external electric circuit to the positive terminal. When a battery is connected to an external electric load, a redox reaction converts high-energy reactants to lower-energy products, and the free-energy difference is delivered to the external circuit as electrical energy. Historically the term "battery" specifically referred to a device composed of multiple cells; however, the usage has evolved to include devices composed of a single cell.

Primary (single-use or "disposable") batteries are used once and discarded, as the electrode materials are irreversibly changed during discharge; a common example is the alkaline battery used for flashlights and a multitude of portable electronic devices. Secondary (rechargeable) batteries can be discharged and recharged multiple times using an applied electric current; the original composition of the electrodes can be restored by reverse current. Examples include the lead–acid batteries used in vehicles and lithium-ion batteries used for portable electronics such as laptops and mobile phones.

Batteries come in many shapes and sizes, from miniature cells used to power hearing aids and wristwatches to, at the largest extreme, huge battery banks the size of rooms that provide standby or emergency power for telephone exchanges and computer data centers. Batteries have much lower specific energy (energy per unit mass) than common fuels such as gasoline. In automobiles, this is somewhat offset by the higher efficiency of electric motors in converting electrical energy to mechanical work, compared to combustion engines.



Fig 3.12 Electric Battery

3.11 PUSH BUTTON



Fig 3.13 Push Button

3.11.1 Description

Push-Buttons are normally-open **tactile switches**. Push buttons allow us to power the circuit or make any particular connection only when we press the button. Simply, it makes the circuit connected when pressed and breaks when released. A push button is also used for triggering of the SCR by gate terminal. These are the most common buttons which we see in our daily life electronic equipment's. Some of the applications of the Push button are mentioned at the end of the article.

3.11.2 Features

- Prevent flux rise by the insert-molded terminal
- Snap-in mount terminal
- Contact Bounce: MAX 5mS
- Crisp clicking by tactile feedback

Dielectric Withstanding Voltage 250V AC for 1 minute

3.12 RESISTORS

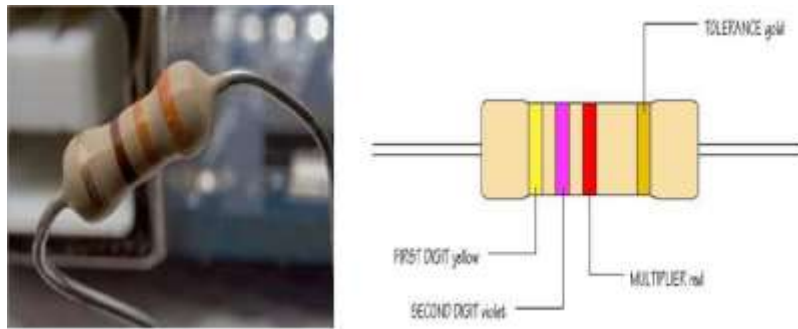


Fig 3.14 Carbon Film Resistor

A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element. The current through a resistor is in direct proportion to the voltage across the resistor's terminals. This relationship is represented by magnitude. When specifying that resistance in an electronic design, the required precision of the resistance may require attention to the manufacturing tolerance of the chosen resistor, according to its specific application. The temperature coefficient of the resistance may also be of concern in some precision applications. Practical resistors are also specified as having a maximum power rating which must exceed the anticipated power dissipation of that resistor in a particular circuit: this is mainly of concern in power electronics applications. Resistors with higher power ratings are physically larger and may require heatsinks. In a high-voltage circuit, attention must sometimes be paid to the rated maximum working voltage of the resistor. While there is no minimum working voltage for a given resistor, failure to account for a resistor's maximum rating may cause the resistor to incinerate when current is run through it.

3.13 VOLTAGE SENSOR

3.13.1 Description

In an electric vehicle Battery Management System (BMS), voltage sensors serve as fundamental components for monitoring and managing the voltage levels across the battery cells and packs.



Fig 3.15 Voltage Sensor

Voltage sensors play a crucial role in continuously measuring the voltage of individual battery cells or modules within the battery pack. This monitoring is essential for assessing the state of charge (SoC) and state of health (SoH) of the battery, as well as detecting any imbalances or abnormalities in voltage levels between cells.

Moreover, voltage sensors contribute significantly to cell balancing, a process aimed at equalizing the voltage levels of individual cells within the battery pack. By detecting deviations in voltage, the BMS can activate balancing circuits or systems to redistribute energy among cells, ensuring optimal performance and longevity of the battery pack.

These sensors also assist in preventing overvoltage and undervoltage conditions that could lead to cell degradation or safety hazards. If any cell voltage exceeds or drops below predefined thresholds, the BMS can trigger protective measures such as cell isolation or active cooling to mitigate the risk of overcharging or deep discharge.

Additionally, voltage sensors play a crucial role in fault detection by continuously monitoring voltage levels and comparing them against predefined thresholds. Any deviations from expected voltage ranges can indicate potential faults or abnormalities within the battery system, prompting the BMS to take corrective actions or issue diagnostics alerts.

Furthermore, voltage sensors provide essential data for logging and reporting purposes. They record voltage measurements over time, allowing the BMS to track the historical performance and behavior of the battery pack. This data is valuable for diagnostics, performance optimization, and predictive maintenance strategies.

Finally, voltage sensor data is integrated into BMS algorithms for state estimation, prediction, and control purposes. By combining voltage measurements with other sensor data such as temperature and current, the BMS can accurately assess the overall health and behavior of the battery system, enabling efficient energy management and optimized vehicle performance.

In summary, voltage sensors are integral components of electric vehicle BMS projects, facilitating precise voltage monitoring, cell balancing, protection against overvoltage and undervoltage conditions, fault detection, data logging, and integration with BMS algorithms. Their role is essential for ensuring the safety, reliability, and longevity of the battery system in electric vehicles.

3.14 CURRENT SENSORS



Fig 3.16 Current Sensor

A current sensor is a device that detects and converts current to an easily measurable output voltage, which is proportional to the current through the measured path. There are a wide variety of sensors, and each sensor is suitable for a specific current range and environmental condition. Among these sensors, a current sensing resistor is the most commonly used. It can be considered a current-to-voltage converter, where inserting a resistor into the current path, the current is converted to voltage in a linear way. The technology used by the current sensor is important because different sensors can have different characteristics for a variety of applications.

Current sensors are based on either open or closed loop hall effect technology. A closed-loop sensor has a coil that is actively driven to produce a magnetic field that opposes the field produced by the

current being sensed. The hall sensor is used as a null-detecting device, and the output signal is proportional to the current being driven into the coil, which is proportional to the current being measured.

In an open loop current sensor, the magnetic flux created by the primary current is concentrated in a magnetic circuit and measured using a hall device. The output from the hall device is the signal conditioned to provide an exact (instantaneous) representation of the primary current.

3.15 THERMISTOR NTC 103

In the domain of electric vehicle Battery Management Systems (BMS), the NTC103 thermistor emerges as a key player, offering precise temperature sensing capabilities crucial for maintaining battery health and safety. As a Negative Temperature Coefficient (NTC) thermistor, the NTC103 exhibits a characteristic decrease in resistance as temperature rises. This unique property allows it to accurately measure temperature variations within the battery pack, offering insights into thermal conditions critical for optimal battery operation.



Fig 3.17 Thermistor

The primary role of the NTC103 thermistor within the BMS project is temperature sensing, where it serves as the frontline sensor for monitoring the thermal landscape of battery cells or modules. By strategically placing NTC103 thermistors throughout the battery pack, the BMS can capture temperature fluctuations across different areas, enabling a comprehensive understanding of the pack's thermal behaviour. This real-time temperature data serves as the foundation for implementing effective thermal management strategies tailored to maintain optimal operating temperatures and prevent overheating-related issues.

Furthermore, temperature monitoring facilitated by NTC103 thermistors plays a pivotal role in ensuring battery safety within the BMS. Elevated temperatures can serve as early indicators of potential hazards such as overcharging, excessive current draw, or internal short circuits. By continuously monitoring temperature levels, the BMS can swiftly detect abnormal thermal patterns and implement proactive measures to mitigate risks. These measures may include adjusting charging rates, activating cooling systems, or initiating emergency shutdown procedures to prevent catastrophic failures and safeguard user safety.

Beyond real-time monitoring, the NTC103 thermistor enables the BMS to engage in data logging and analysis, offering valuable insights for diagnostics and optimization. By logging temperature data over time, the BMS can track historical trends, identify potential failure modes, and fine-tune thermal management strategies for enhanced efficiency and reliability. This data-driven approach empowers the BMS to make informed decisions, predict potential issues before they escalate, and optimize battery performance and lifespan over the long term.

In essence, the NTC103 thermistor serves as the eyes and ears of the electric vehicle BMS, providing crucial temperature data essential for ensuring battery health, safety, and performance. Its accurate

sensing capabilities, coupled with the BMS's proactive response mechanisms, form a robust framework for effective thermal management and safeguarding the integrity of the battery system.

3.16 I2C MODULE:



Fig 3.18 I2C Module

3.16.1 Description

I2C Module has a inbuilt PCF8574 I2C chip that converts I2C serial data to parallel data for the LCD display.

These modules are currently supplied with a default I2C address of either 0x27 or 0x3F. To determine which version you have check the black I2C adaptor board on the underside of the module. If there a 3 sets of pads labeled A0, A1, & A2 then the default address will be 0x3F. If there are no pads the default address will be 0x27.

The module has a contrast adjustment pot on the underside of the display. This may require adjusting for the screen to display text correctly.

3.16.2 Features

- Operating Voltage: 5V
- Backlight and Contrast is adjusted by potentiometer
- Serial I2C control of LCD display using PCF8574
- Come with 2 IIC interface, which can be connected by Dupont Line or IIC dedicated cable
- Compatible for 16x2 LCD
- This is another great IIC/I2C/TWI/SPI Serial Interface
- With this I2C interface module, you will be able to realize data display via only 2 wires.

3.17 LCD DISPLAY

3.17.1 LCD 16×2 Pin Configuration and Its Working

Nowadays, we always use the devices which are made up of LCDs such as CD players, DVD players, digital watches, computers, etc. These are commonly used in the screen industries to replace the utilization of CRTs. Cathode Ray Tubes use huge power when compared with LCDs, and CRTs

heavier as well as bigger. These devices are thinner as well power consumption is extremely less. The LCD 16×2 working principle is, it blocks the light rather than dissipate. This article discusses an overview of LCD 16X2, pin configuration and its working.

3.17.2 What is the LCD 16×2?

The term LCD stands for liquid crystal display. It is one kind of electronic display module used in an extensive range of applications like various circuits & devices like mobile phones, calculators, computers, TV sets, etc. These displays are mainly preferred for multi-segment light-emitting diodes and seven segments. The main benefits of using this module are inexpensive; simply programmable, animations, and there are no limitations for displaying custom characters, special and even animations, etc.



Fig 3.19 16X2 LCD

The 16×2 LCD pinout is shown in fig 4.12.

- Pin1 (Ground/Source Pin): This is a GND pin of display, used to connect the GND terminal of the microcontroller unit or power source.
- Pin2 (VCC/Source Pin): This is the voltage supply pin of the display, used to connect the supply pin of the power source.
- Pin3 (V0/VEE/Control Pin): This pin regulates the difference of the display, used to connect a changeable POT that can supply 0 to 5V.
- Pin4 (Register Select/Control Pin): This pin toggles among command or data register, used to connect a microcontroller unit pin and obtains either 0 or 1 (0 = data mode, and 1 = command mode).
- Pin5 (Read/Write/Control Pin): This pin toggles the display among the read or writes operation, and it is connected to a microcontroller unit pin to get either 0 or 1 (0 = Write Operation, and 1 = Read Operation).
- Pin 6 (Enable/Control Pin): This pin should be held high to execute Read/Write process, and it is connected to the microcontroller unit & constantly held high.
- Pins 7-14 (Data Pins): These pins are used to send data to the display. These pins are connected in two-wire modes like 4-wire mode and 8-wire mode. In 4-wire mode, only four pins are connected to the microcontroller unit like 0 to 3, whereas in 8-wire mode, 8-pins are connected to microcontroller unit like 0 to 7.
- Pin15 (+ve pin of the LED): This pin is connected to +5V
- Pin 16 (-ve pin of the LED): This pin is connected to GND.



Fig 3.20 LCD-16×2-pin-diagram

CHAPTER – 4

4.1 INTRODUCTION

The implementation of software within the context of this electric vehicle Battery Management System (BMS) project is integral to its success, as it orchestrates the monitoring, analysis, and control of various critical parameters essential for battery health and vehicle performance. Leveraging sophisticated algorithms and real-time data processing capabilities, the software enables seamless communication between hardware components, facilitates precise measurement of voltage, current, and temperature, and implements proactive measures to ensure the safety and efficiency of the battery system.

In this chapter, we delve into the intricacies of the software architecture, detailing its design principles, functionalities, and performance metrics. Furthermore, we present the results of our implementation, showcasing its efficacy in optimizing battery management, enhancing operational safety, and achieving optimal performance in electric vehicle applications. Through comprehensive analysis and evaluation, we aim to provide insights into the significance of software in shaping the future of electric vehicle technology and advancing sustainable mobility solutions.

4.2 OPERATION

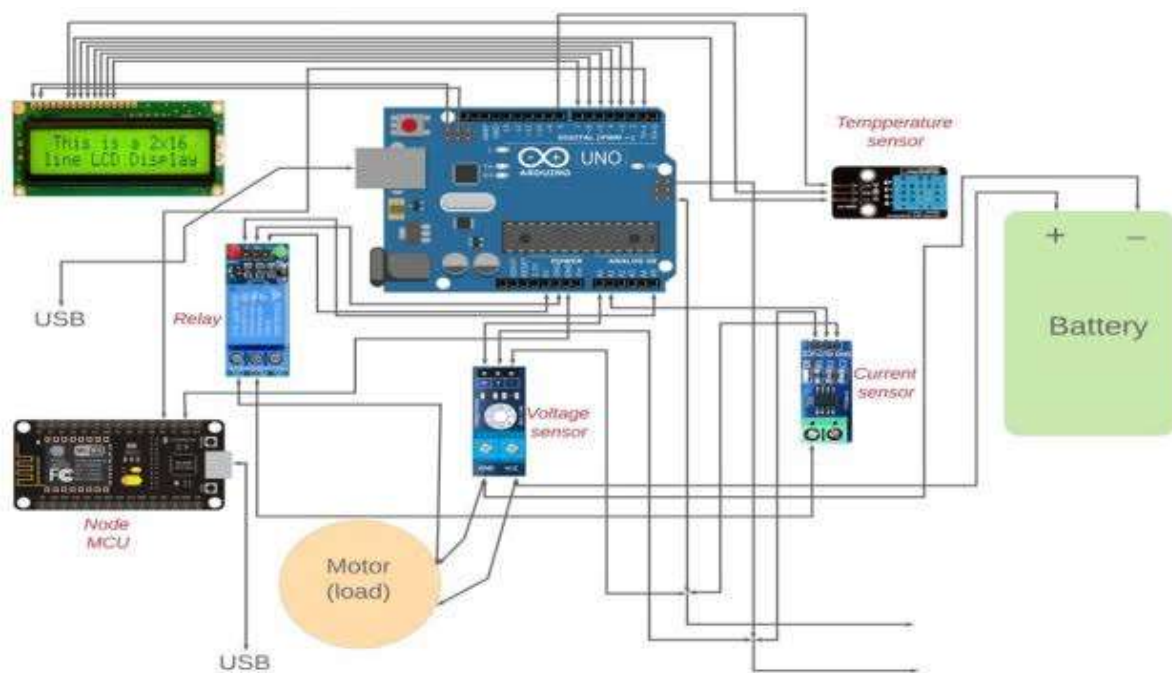


Fig 4.1 Circuit Diagram

The Electric Vehicle Battery Management System (BMS) and Fire Protection project integrates advanced technologies to ensure the safe and efficient operation of electric vehicle (EV) battery systems. The project begins with the continuous monitoring of crucial parameters such as voltage, temperature, and current of each battery cell within the pack. This monitoring is facilitated by sensors strategically placed throughout the battery pack. The BMS utilizes this data to detect any abnormalities or potential risks to the battery system.

One of the key functions of the BMS is cell balancing, which ensures that all battery cells within the pack are charged and discharged uniformly. By implementing cell balancing algorithms, the BMS can regulate the charging and discharging rates of individual cells, maximizing battery capacity and longevity. Moreover, the BMS controls the charging and discharging processes to prevent overcharging, over discharging, or overheating of the battery pack. It communicates with the vehicle's charging system to optimize charging parameters based on battery condition and environmental factors.

In addition to monitoring and managing battery performance, the BMS is equipped with safety features to mitigate potential hazards. It continuously monitors the battery pack for signs of safety risks such as thermal runaway or internal shorts. Upon detection of a safety hazard, the BMS can trigger various safety measures, including reducing charging/discharging rates, isolating faulty cells, or shutting down the battery pack altogether.

Furthermore, the project incorporates a fire detection and suppression system to enhance safety measures. Smoke and heat sensors are strategically placed within the battery pack and surrounding areas to detect signs of fire hazards. Upon detecting a fire hazard, the system activates the fire suppression mechanism, which may involve releasing fire retardants, triggering isolation switches, or activating other safety protocols to contain the fire and prevent its spread.

The integration of the BMS and fire protection systems with the vehicle's main control unit enables real-time monitoring and control of battery and safety systems. In case of emergencies such as fire or battery failure, the vehicle's control unit can initiate appropriate responses, such as activating safety protocols, alerting the driver, or shutting down critical systems to prevent further damage. Overall, the project aims to enhance the reliability and safety of electric vehicles, contributing to their widespread adoption and acceptance.

4.3 SYSTEM DESCRIPTION

We can check the Temperature sensor data along with the battery voltage and battery percentage in both the charging and discharging on both cases. They show smartphones or Computer dashboards from anywhere in the world

4.3.1 Voltage Detection

The cut-off voltage of the battery is 2.8V, while the battery's maximum voltage is 25V. The Arduino Nano Analogue Pin will successfully support any voltage below 3.3V. We must first descend from the higher voltage level. Two resistors are present, and the supply voltage is 4.2V. A result of 2.1V will occur from this. The base value is 2.8V in essence, and the cut-off voltage drops down to 1.4V using the same voltage divider organisation. As a result, the Arduino Nano Analogue Pin maintains both the higher and lower voltage. If the voltage rises beyond 4.2V, the automated supply of power will be cut off.

4.3.2 Voltage Divider Network Calculations

The cut-off voltage of the battery is 2.8V, while the battery's maximum voltage is 25V. The Arduino Analogue Pin can readily accommodate any voltage lower than 3.3V. We must first descend to the higher voltage level.

Two 100K resistors are present, and the supply voltage is 4.2V. A result of 2.1V will occur from this. The base voltage is also 2.8V, and the cut-off voltage drops down to 1.4V using the same voltage divider structure. Consequently, the Arduino Analogue Pin maintains both the upper and lower voltage.

4.3.3 Temperature Detection

We will plan a framework to screen NTC 103 temperature, battery voltage, alongside charging and releasing status. For the microcontroller, we use Arduino and NodeMCU, which has an ESP8266 Wi-Fi-empowered chip. You can likewise utilize a more modest board like the Wemos D1 Mini. This Wi-Fi chip can associate with the Wi-Fi organization and routinely transfer the information to the server. The Out pin of the NTC 103 sensor is associated with the digital pin of the Arduino. SCL and SDA pins are associated with D1 and D2 pins. Though VCC and GND of DHT11

4.3.4 Calculation of Battery Percentage

We show the battery percentage on the Blynk template dashboard and show it on the mobile screen. Battery percentage showing in between 1 to 100. special feature is Blynk IoT user can access the data directly anywhere in the world. The mechanized Battery Monitoring System vehicle with different control components can be worked by making associations. For the microcontroller, we utilize the Arduino and NodeMCU, which has an ESP8266 WIFI-empowered chip. You can likewise utilize more modest sheets like the Wemos D1 Mini. This Wi-Fi chip can interface with a Wi-Fi organization and transfer information to the server routinely.

4.3.5 Current Calculation

The Hall sensor is used to measure the current flow in the circuit. The Allegro™ ACS712 is a Hall sensor imparting in your price range and unique answer in industries, business and verbal exchange structures. The hall sensor is utilised to gauge current flow in the cable. The hall sensor can gauge the current by applying a rigid resistance to the wire. In order to pass past the corridor sensor, a portion of the wire from the transistor that is heading to the appliance is cut down. Three pins make up the corridor sensor: a voltage pin, a ground pin, and an output pin. The output pin is connected to the analogue pin within the Arduino, and the voltage pin is connected to the 5-volt supply from the Arduino. The floor pin is connected to the ground. The Arduino pin analog is connected to the analogue pin A2 of the hall sensor, which measures the current in the fan, and the analogue pin A2 of the light sensor, which measures the current in the light.

4.4 SOFTWARE

For the software we use the Blynk IoT Platform. In this platform first you need create account, after creating account define the parameters what you want. In this module we designed for measure the temperature, voltage, current and presence of gases.

4.4.1 CONFIGURING THE BLYNK IOT DASHBOARD

Setting up the Blynk IoT Cloud dashboard is required before you can monitor sensor and battery data on the Blynk IoT Server. Please go to <https://blynk.cloud/> to configure the Blynk Server. Create a new account or sign in if one already exists.

4.4.2 Making the Blynk Template

A template is a task that allows us to build a web and mobile dashboard for certain hardware. It is an IoT smart device in our scenario. You need to select the New Template in order to start a project.

- Enter a template name. I am Giving it “Smart IoT”.
- Select the hardware board (ESP8266).
- The connection type will be Wi-Fi.
- You can add a description of your project. If required
- Select "Done."
- The template has now been made.

You may then select a category based on your project after that. Don't worry if your category isn't on the list; you may select another. A temperature sensor is what I'm choosing.

4.4.3 Making Events for Notification Alerts on Blynk

Systems for notifying and alerting users use events. I'm making events now to track Battery Percentage. A notice is delivered to your mobile phone and an event is triggered if the battery percentage falls below the threshold level.

- Press the add event button.
- Enter the event Name. For me, it's “Battery Low “
- Choose your event color. I am selecting Red.
- Select the type of event. I am choosing Critical.
- Enter the description of your event.
- Now go to the notification tab then enable notification.
- Then select push notification to device owner every 1 minute.
- Click on save

4.5 RESULTS

In Blynk app, it is shown that the Battery Voltage, Current, Power and Temperature.

Case I : No Charging

Case II : Slow Charging Mode

Case III : Fast Charging Mode



Fig 4.2(a) Case I

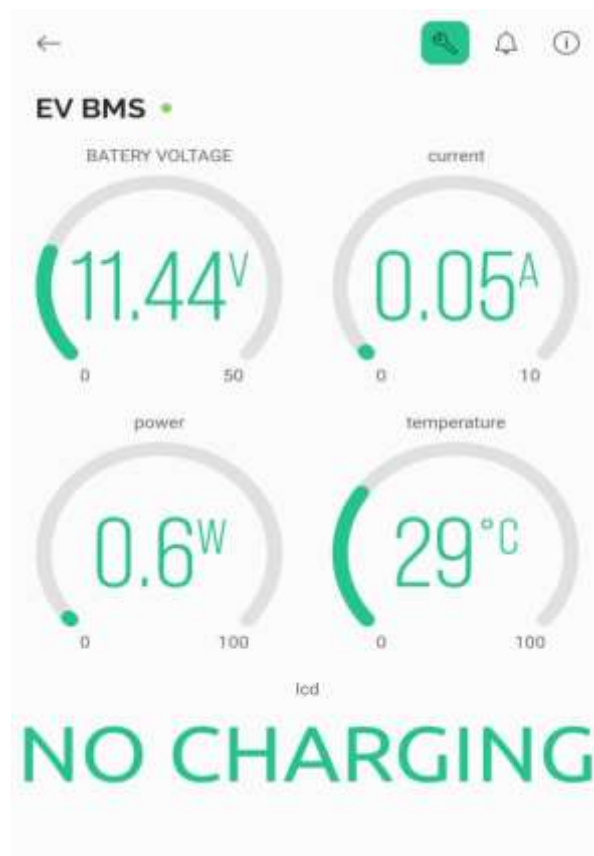


Fig 4.2(b) Case I

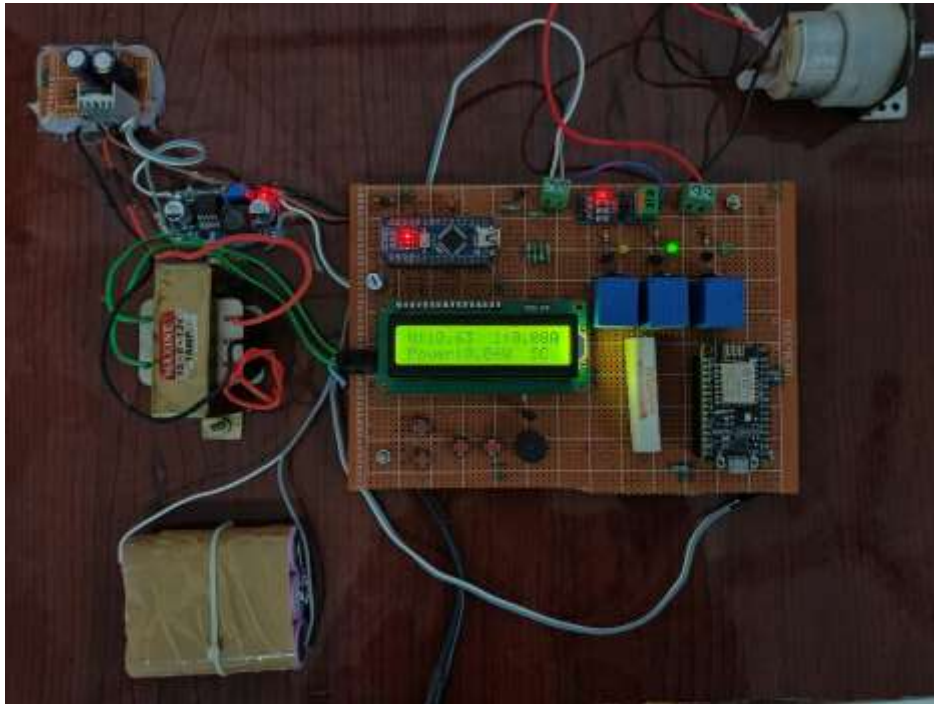


Fig 4.3(a) Case II



Fig 4.3(b) Case II

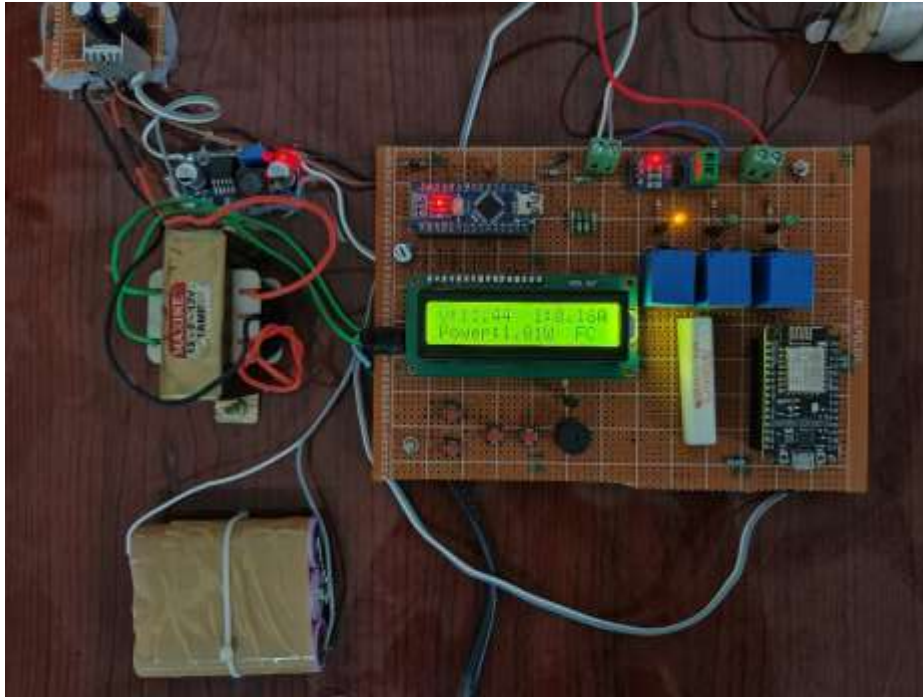


Fig 4.4(a) Case III

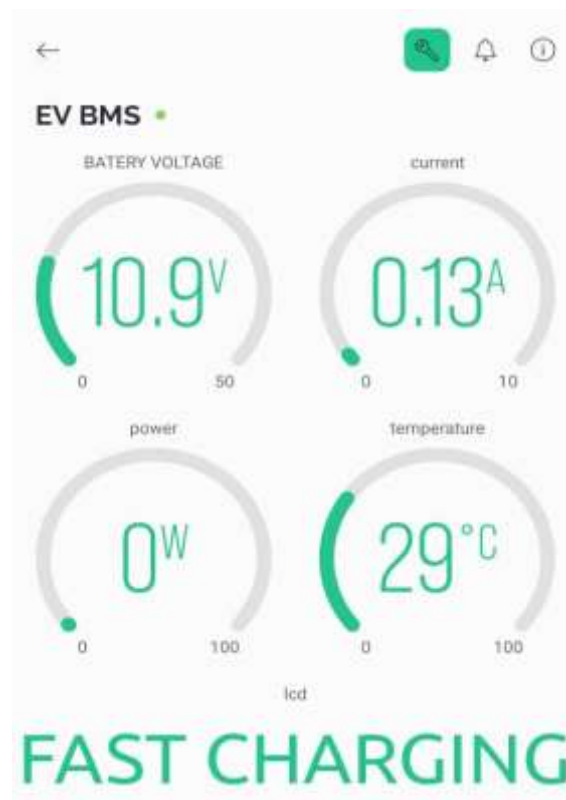


Fig 4.4(b) Case III

CHAPTER – 5

5.1 CONCLUSION

In conclusion, this electric vehicle Battery Management System (BMS) project represents a significant advancement in the field of electric vehicle technology. Through the integration of sophisticated software, precise sensors, and intelligent algorithms, the BMS demonstrates its capability to monitor, manage, and optimize the performance of battery packs in electric vehicles.

It is very important for BMS to well-maintained the battery reliability and safety, the state monitoring and evaluation, cell balancing and charge control are well functional. Thus, this present paper is review on BMS, focussing study for optimization of BMS that will lead to reliability of BMS and optimize power performance. Monitor the battery anywhere in the world using IoT. It's taking the data from IoT embedded sensor and transmit the cloud. Provide better performance to user enhance the battery life. This IoT based battery management system help us to monitor the battery conditions and also help us to monitor the battery health.

The software architecture, meticulously designed and implemented, orchestrates the seamless communication between hardware components, enabling real-time monitoring of critical parameters such as voltage, current, and temperature. Leveraging advanced algorithms, the BMS accurately estimates state of charge (SoC), predicts potential faults, and implements proactive measures to ensure the safety and efficiency of the battery system.

Furthermore, the results obtained from the implementation of the BMS project underscore its efficacy in enhancing battery management, improving operational safety, and maximizing vehicle performance. Through comprehensive analysis and evaluation, the project demonstrates the significance of software-driven solutions in shaping the future of electric vehicle technology and advancing sustainable mobility solutions.

5.2 FURTHER WORK

We think about fully system is make as AI. We want to made an AI Based Battery management system. In this AI fully system is controlled by AI. Few things only handle by human hand. While the current project marks a significant milestone in electric vehicle BMS development, there are several avenues for further research and improvement:

Integration of Advanced Algorithms: Explore the integration of machine learning and artificial intelligence techniques for enhanced battery management, fault detection, and predictive maintenance.

Enhanced Sensor Technologies: Investigate the integration of advanced sensor technologies, such as fiber optic sensors and wireless sensor networks, for real-time monitoring of battery parameters and environmental conditions.

Optimization of Thermal Management: Further optimize thermal management strategies to ensure efficient cooling and heating of the battery pack, thereby enhancing overall battery performance and lifespan.

Implementation of Redundancy and Fault Tolerance: Explore the implementation of redundant sensor and communication systems to improve fault tolerance and reliability, ensuring continuous operation even in the event of sensor or component failures.

Validation and Field Testing: Conduct extensive validation and field testing of the BMS in real-world electric vehicle applications to assess its performance, reliability, and scalability under various operating conditions.

By pursuing these avenues for further work, the project aims to continue advancing electric vehicle technology, driving innovation, and contributing to the development of sustainable mobility solutions for the future.

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APPENDIX

CODE

NODE MCU

```
//////////node mcu side

#define BLYNK_TEMPLATE_ID "TMPL3GhI-_DFf"

#define BLYNK_TEMPLATE_NAME "EV BMS"

#define BLYNK_AUTH_TOKEN "-G5z3zuywVIZPwYBqpwemN8Fsm-NhYUZ"


#define BLYNK_PRINT Serial

#include <ESP8266WiFi.h>

#include <BlynkSimpleEsp8266.h>

char auth[] = BLYNK_AUTH_TOKEN;

char ssid[] = "Nagu"; // Enter your Wifi Username

char pass[] = "nagu@213"; // Enter your Wifi password

SimpleTimer timer;

int s_charge = D3;

int F_charge = D2;

int firstVal;

int secondvalue;

String myString; // complete message from arduino, which consists of snesors data


char rdata; // received charactors

String data;

String I;

String J;

String K;

String m;

String n;

bool myflag = false;

bool myflag1 = false;
```

```
void myTimerEvent()
{
// You can send any value at any time.
// Please don't send more that 10 values per second.
Blynk.virtualWrite(V1, millis() / 1000);
}

void setup()
{
// Debug console
Serial.begin(115200);
pinMode(s_charge,INPUT);
pinMode(F_charge,INPUT);
Blynk.begin(auth, ssid, pass, "blynk.cloud", 80);
timer.setInterval(1000L,sensorvalue);
timer.setInterval(1000L,sensorvalue1);
timer.setInterval(1000L,sensorvalue2);
timer.setInterval(1000L,sensorvalue3);
timer.setInterval(1000L,sensorvalue4);
timer.setInterval(1000L,sensorvalue5);
timer.setInterval(1000L,sensorvalue6);
}

void loop()
{
if (Serial.available() == 0 )
{
Blynk.run();
timer.run(); // Initiates BlynkTimer
}
if (Serial.available() > 0 )
{
```

```
rdata = Serial.read();  
myString = myString+ rdata;  
if( rdata == '\n')  
{  
  
    I = getValue(myString, ',', 0);  
    J = getValue(myString, ',', 1);  
    K = getValue(myString, ',', 2);  
    m = getValue(myString, ',', 3);  
    n = getValue(myString, ',', 4);  
  
    //m = myString.toInt();  
  
    myString = " ";  
    // end new code  
    }  
    }  
  
}
```

```
void sensorvalue(){  
    data = data + I;  
    Blynk.virtualWrite(V0, I);  
    Serial.println(data);  
  
    data = "";  
}  
void sensorvalue1(){  
    data = data + J;
```

```
Blynk.virtualWrite(V4,J);  
Serial.println(data);  
data = "";  
  
}  
  
void sensorvalue2(){  
    data = data + K;  
  
    Blynk.virtualWrite(V2, K );  
    Serial.println(data);  
    data = "";  
}  
  
void sensorvalue3(){  
  
    data = data + firstVal;  
    Blynk.virtualWrite(V5,firstVal );  
    Serial.println(data);  
  
if(firstVal >=40 && myflag ==true){  
  
    Blynk.logEvent("temp");  
    Blynk.virtualWrite(V3, "Temperature above 40 c");  
    myflag = false;  
  
}if(firstVal <40 ){  
  
    Blynk.virtualWrite(V3, "Temperature below 40 c");  
    myflag = true;  
    delay(1200);
```

```
}

    data = "";
}void sensorvalue4(){
    data = data + n;

    Blynk.virtualWrite(V6, n);
    Serial.println(data);

    data = "";
}

void sensorvalue5(){

int s_charge_state = digitalRead(s_charge);
int F_charge_state = digitalRead(F_charge);
    if( s_charge_state ==0){
        Blynk.virtualWrite(V3, "slow CHARGING");
    }
    if( F_charge_state ==0){
        Blynk.virtualWrite(V3, "fast CHARGING");
    }
    if( s_charge_state ==1 && F_charge_state ==1 ){
        Blynk.virtualWrite(V3, "NO CHARGING");
        /*Serial.print("s_charge_state :");
Serial.println(s_charge_state );
Serial.print("F_charge_state :");
Serial.println(F_charge_state );
delay(500);*/
    }
```

```
}  
  
void sensorvalue6(){  
    data = data + secondvalue;  
  
    Serial.println(secondvalue);  
  
    if(secondvalue >=12.5 && myflag1 == false){  
        Serial.println("battery fully charged");  
        Blynk.logEvent("batteryfull");  
        myflag1 = true;  
    }if(secondvalue <=5 && myflag1 == true){  
        Serial.println("battery LOW ");  
        Blynk.logEvent("batterylow");  
        myflag1 = false;  
    }  
  
    data = "";  
}  
  
String getValue(String data, char separator, int index)  
{  
    int found = 0;  
    int strIndex[] = { 0, -1 };  
    int maxIndex = data.length() - 1;  
  
    for (int i = 0; i <= maxIndex && found <= index; i++) {  
        if (data.charAt(i) == separator || i == maxIndex) {  
            found++;  
            strIndex[0] = strIndex[1] + 1;  
            strIndex[1] = (i == maxIndex) ? i+1 : i;  
        }  
    }  
    return found > index ? "" : data.substring(strIndex[0], strIndex[1]);  
}
```

```
}  
}  
return found > index ? data.substring(strIndex[0], strIndex[1]) : "";  
}
```

ARDUINO

```
//////////arduino side  
#include <LiquidCrystal.h>  
const int rs = 2, en = 3, d4 = 4, d5 = 5, d6 = 6, d7 = 7;  
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);  
#include <SoftwareSerial.h>  
SoftwareSerial nodemcu(9,8);  
int bat_voltage = A1;  
#define temperature_sensor A6 //the thermistor attach to  
#define beta 3950 //the beta of the thermistor  
#define resistance 10 //the value of the pull-downresistor  
#define up 10  
#define down 11  
#define enter 12  
#define Clear A0  
byte pointer[]={0x00,0x00,0x04,0x02,0x1D,0X02,0X04,0X00};  
int downsel=0;  
int F_relay = A4;  
int S_relay = A3;  
int LOAD_relay = A5;  
int buzzer = 13;  
float bat_vin;  
float bat_vout;  
float R1 = 100000.0;//V IN  
float R2 = 10000.0;//V ARDUINO
```

```
int b_value =0;
const int currentPin = A2;
int sensitivity2 = 185;
int bat_value = 0;
int offsetVoltage2 = 2500;
double adcVoltage2 = 0;
double currentValue = 0.0;
float bat_power;
float tempC;

String sensor1;
String sensor2;
String sensor3;
String sensor4;
String sensor5;
String sensor6;
String sensor7;
String sensor8;
String sensor9;
String cdata ="";
bool myflag = false;
void setup() {
  Serial.begin(9600);
  nodemcu.begin(115200);
  lcd.begin(16,2);
  pinMode(bat_voltage, INPUT);

  pinMode(up,INPUT_PULLUP);
  pinMode(down,INPUT_PULLUP);
  pinMode(enter,INPUT_PULLUP);
```



```
pinMode(Clear,INPUT_PULLUP);

pinMode(F_relay,OUTPUT);
pinMode(S_relay,OUTPUT);
pinMode(LOAD_relay,OUTPUT);
pinMode(buzzer,OUTPUT);

digitalWrite(buzzer,LOW);
lcd.setCursor(3,0);
lcd.print("Hello, world!");// Print a message to the LCD.

delay(1000);
lcd.clear();

lcd.createChar(5, pointer);
}
void loop() {
  UPandDOWN();
  parameters();
}
void parameters(){
  b_value = analogRead(bat_voltage);
  bat_vout = (b_value * 5.0) / 1024.0;
  bat_vin = bat_vout / (R2 / (R1 + R2));
  if ( bat_vin < 0.09)
  {
    bat_vin= 0.0;
  }
  if ( bat_vin >= 13.5)
  { downsel =0;
```

```
    delay(2500);
    downsel =6;
}if ( bat_vin <= 5)
{
    downsel =3;
}

bat_value = analogRead(currentPin);
adcVoltage2 = (bat_value / 1024.0) * 5000;
currentValue = ((adcVoltage2 - offsetVoltage2) / sensitivity2);
long a = analogRead(temperature_sensor);
tempC = beta /(log((1024.0 * 10 / a - 10) / 10) + beta / 298.0) - 273.0;
bat_power = (bat_vin * currentValue);

if(tempC>=45.00 ){
    downsel =6;
    lcd.clear();
    lcd.setCursor(1,0);
    lcd.print("Battery ");
    lcd.setCursor(0,1);
    lcd.print("temperature high");
    digitalWrite(buzzer,HIGH);
    delay(500);
    digitalWrite(buzzer,LOW);
    delay(500);
    digitalWrite(buzzer,HIGH);
    delay(500);
    digitalWrite(buzzer,LOW);
    delay(500);
    digitalWrite(buzzer,HIGH);
```

```
    delay(500);
    digitalWrite(buzzer,LOW);
    delay(500);
    digitalWrite(buzzer,HIGH);
    delay(500);
    digitalWrite(buzzer,LOW);
    delay(500);
    digitalWrite(buzzer,HIGH);
    delay(500);
    digitalWrite(buzzer,LOW);
    delay(500);
}

sensor1 = bat_vin ;
sensor2 = currentValue;
sensor3 = bat_power;
sensor4 = tempC;
sensor5 = 84.6 ;

cdata = cdata + sensor1 + "," + sensor2 + "," + sensor3 + "," + sensor4 + "," + sensor5 ; //
comma will be used a delimiter

Serial.println(cdata);
nodemcu.println(cdata);
cdata = " ";
delay(100);
}

void UPandDOWN(){
    lcd.clear();
    switch(downdsel){
    case 0: lcd.setCursor(0,0);
    lcd.write(5);
    lcd.setCursor(1,0);
    lcd.print("BATTERY ");
```

```
break;  
case 1: lcd.setCursor(0,0);  
lcd.write(5);  
lcd.setCursor(1,0);  
lcd.print("TEMPERATURE");  
break;
```

```
case 2: lcd.setCursor(0,0);  
lcd.write(5);  
lcd.setCursor(1,0);  
lcd.print("FAST CHARGE");  
break;
```

```
case 3: lcd.setCursor(0,0);  
lcd.write(5);  
lcd.setCursor(1,0);  
lcd.print("SLOW CHARGE");  
break;
```

```
case 4: lcd.setCursor(0,0);  
lcd.write(5);  
lcd.setCursor(1,0);  
lcd.print("DISHCHARGE ON");  
break;
```

```
case 5: lcd.setCursor(0,0);  
lcd.write(5);  
lcd.setCursor(1,0);  
lcd.print("DISHCHARGE OFF");  
break;
```

```
case 6: lcd.setCursor(0,0);  
lcd.write(5);  
lcd.setCursor(1,0);  
lcd.print("CHARGER OFF");  
break;  
}
```

```
if(digitalRead(down) == LOW){  
    downsel++;  
    Serial.println(downsel);  
    if( downsel>=7){  
        downsel=0;  
    }  
    delay(100);  
}
```

```
if(digitalRead(up)== LOW){  
    downsel--;  
    if( downsel<=-1){  
        downsel=6;  
    }  
    Serial.println(downsel);  
    delay(100);  
}
```

```
if(digitalRead(enter)== LOW){  
    while(1){  
        ENTER();  
        parameters();  
    }  
if(digitalRead(Clear)== LOW){  
    while(1){
```

```
UPandDOWN();  
}  
}  
}  
}  
delay(100);  
}  
void ENTER(){  
    switch(downsel){  
case 0:  
    lcd.clear();  
    lcd.setCursor(0,0);  
    lcd.print("V:");  
    lcd.print(bat_vin);  
    lcd.setCursor(9,0);  
    lcd.print("I:");  
    lcd.print(currentValue);  
    lcd.print("A");  
    lcd.setCursor(0,1);  
    lcd.print("Power:");  
    lcd.print(bat_power);  
    lcd.print("W");  
    delay(150);  
    break;  
  
case 1:  
    lcd.clear();  
    lcd.setCursor(0,0);  
    lcd.print("TEMP STATUS");  
    lcd.setCursor(0,1);
```

```
lcd.print("TEMP:");  
lcd.print( tempC);  
lcd.print("C");  
delay(150);  
break;  
  
case 2:  
lcd.clear();  
digitalWrite(S_relay,HIGH);  
digitalWrite(F_relay,LOW);  
digitalWrite(LOAD_relay,LOW);  
lcd.setCursor(0,0);  
lcd.print("V:");  
lcd.print(bat_vin);  
lcd.setCursor(9,0);  
lcd.print("I:");  
lcd.print(currentValue);  
lcd.print("A");  
lcd.setCursor(0,1);  
lcd.print("Power:");  
lcd.print(bat_power);  
lcd.print("W");  
lcd.setCursor(13,1);  
lcd.print("FC");  
delay(1000);  
lcd.setCursor(13,1);  
lcd.print(" ");  
delay(150);  
break;
```

```
case 3:
lcd.clear();
digitalWrite(S_relay,LOW);
digitalWrite(F_relay,HIGH);
digitalWrite(LOAD_relay,LOW);
lcd.setCursor(0,0);
lcd.print("V:");
lcd.print(bat_vin);
lcd.setCursor(9,0);
lcd.print("I:");
lcd.print(currentValue);
lcd.print("A");
lcd.setCursor(0,1);
lcd.print("Power:");
lcd.print(bat_power);
lcd.print("W");
lcd.setCursor(13,1);
lcd.print("SC");
delay(1000);
lcd.setCursor(13,1);
lcd.print(" ");
delay(150);
break;
```

```
case 4:
lcd.clear();
digitalWrite(LOAD_relay,HIGH);
digitalWrite(S_relay,LOW);
digitalWrite(F_relay,LOW);
lcd.setCursor(0,0);
```



```
lcd.print("V:");  
lcd.print(bat_vin);  
lcd.setCursor(9,0);  
lcd.print("I:");  
lcd.print(currentValue);  
lcd.print("A");  
lcd.setCursor(0,1);  
lcd.print("Power:");  
lcd.print(bat_power);  
lcd.print("W");  
lcd.setCursor(13,1);  
lcd.print("DC");  
delay(1000);  
lcd.setCursor(13,1);  
lcd.print(" ");  
delay(150);  
break;  
  
case 5:  
lcd.clear();  
digitalWrite(Load_relay,LOW);  
digitalWrite(S_relay,LOW);  
digitalWrite(F_relay,LOW);  
lcd.setCursor(0,0);  
lcd.print("LOAD TURN OFF ");  
lcd.setCursor(0,1);  
lcd.print("CHARGER TURN OFF ");  
delay(150);  
break;
```

case 6:

lcd.clear();

digitalWrite(S_relay,LOW);

digitalWrite(F_relay,LOW);

digitalWrite(LOAD_relay,LOW);

lcd.setCursor(0,0);

lcd.print("CHARGER OFF ");

lcd.setCursor(0,1);

lcd.print("LOAD OFF ");

delay(150);

break;

}

}